

VISVESVARAYATECHNOLOGICAL UNIVERSITY

JNANA SANGAMA, BELAGAVI-590018



**Activity Assessment Test3 (AAT3) of Advanced VLSI Design
design-based experiment (Miniproject) using Cadence tool.**

“Design Up / Down Counter”

Submitted In partial fulfilment for the award of degree of

BACHELOR OF ENGINEERING

In

ELECTRONICS AND COMMUNICATION ENGINEERING

Submitted by

T Vikas

3BR22EC172

Under the guidance of

Dr. Abdul Lateef Haroon P S

(Assoc.prof)

BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT

Autonomous Institute under VTU,Belagavi

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**(Recognized by Govt. of Karnataka, approved by AICTE, New Delhi & Affiliated to
Visvesvaraya Technological University, Belagavi)**

"Jnana Gangotri" Campus, No.873/2, Bellary-Hosapete Road, Allipur,

Ballari-583 104 (Karnataka) (India)

08392 – 237100 / 237190, Fax: 08392 – 23719



2025-2026

Introduction

In digital systems, **counters** are fundamental sequential circuits used to count clock pulses, events, or operations. They are widely employed in applications such as digital clocks, frequency dividers, microprocessors, and control systems. A counter can be designed to count in an increasing order (**up-counter**) or decreasing order (**down-counter**) depending on the requirement.

An **Up/Down Counter** is a versatile circuit that can perform both operations based on a control signal. When the control input is logic '1', the counter increments its value with each clock pulse; when it is logic '0', it decrements the value. This bidirectional counting makes it useful in systems such as elevator controllers, digital position tracking, and reversible event counters.

Counters can be implemented as **asynchronous (ripple)** or **synchronous** circuits. In asynchronous counters, flip-flops are triggered in sequence, causing propagation delays. In contrast, synchronous counters use a common clock signal for all flip-flops, ensuring faster and more reliable operation. Hence, in this project, a **4-bit synchronous Up/Down Counter** is designed using **D flip-flops** and combinational logic.

The circuit is modeled using **Verilog HDL** and simulated in the **Cadence tool environment** to verify its functionality. The counter's operation is analyzed for both up and down counting modes, and synthesis is carried out to evaluate design metrics such as **area, timing, and power**.

This project provides a clear understanding of sequential circuit design, clock synchronization, and control logic implementation. It also offers practical exposure to the complete **VLSI design flow**, including RTL design, functional simulation, and synthesis analysis using industry-standard tools.

2. Literature Survey

2.1 Problem Statement

Digital systems frequently require counting operations for tasks like event detection, timing, frequency division, and control logic. Many systems also require **bidirectional counting**, such as elevator controllers, rotary encoder tracking, and reversible timers. Traditional

counters either count only upward or only downward and may suffer from delay or power inefficiency depending on the architecture.

2.2 Problem Solution

To address these challenges, a **4-bit synchronous Up/Down Counter** is designed using Verilog HDL.

The synchronous architecture eliminates propagation delay.

The Up/Down control enables bidirectional counting.

Cadence tools help verify timing, area, and power consumption, ensuring real VLSI-level performance.

2.3 Objective

The main objective of this project is to **design, implement, simulate, and analyze a 4-bit Synchronous Up/Down Counter** using **Verilog HDL** and the **Cadence VLSI design environment**. The project aims to provide practical understanding of sequential circuit design and its realization in the VLSI flow.

The specific objectives are as follows:

1. **To design a 4-bit synchronous Up/Down Counter:**

Develop the logic circuit that can count both upward and downward depending on a control input, using D flip-flops and combinational logic.

2. **To implement the design using Verilog HDL:**

Write synthesizable Verilog code that accurately models the counter behavior and supports both counting modes.

3. **To simulate and verify the functionality:**

Use a Verilog testbench in the Cadence simulator to verify correct counting sequence for both up and down modes under various test conditions.

4. **To perform synthesis and analysis:**

Synthesize the RTL design using Cadence Genus (or RTL Compiler) and evaluate key performance metrics such as **area, timing delay, and power consumption**.

5. **To understand the complete VLSI design flow:**

Gain hands-on experience in modeling, verification, and synthesis of a sequential circuit, which forms the foundation for designing more complex digital systems.

2.4 Literature Review

Sl. No.	Title / Author / Year	Introduction / Objective	Implementation (Methodology)	Advantages	Disadvantages	Key Takeaway / Application
1.	“Design of 4-bit Synchronous Up/Down Counter using Verilog HDL”- EngineersGarage (2024)	Presents a basic up/down counter model using behavioral Verilog.	Implemented using D flip-flops and control signal logic.	Simple, fast simulation; easy to verify.	No synthesis or hardware testing.	Ideal for learning Verilog implementation.
2.	“FPGA Based Up/Down Counter using VHDL” — IJERT (2023)	Describes FPGA implementation using VHDL.	Used Xilinx ISE to synthesize and implement a 4-bit counter.	Hardware realization and timing verified.	Limited to FPGA environment.	Demonstrates FPGA-based sequential design.
3.	“Low Power Synchronous Counter using GDI Technique” — JETIR (2020)	Focuses on reducing power consumption in counters.	Designed using Gate Diffusion Input (GDI) logic in 90nm CMOS.	Low power and reduced transistor count.	Complex biasing and limited voltage swing.	Suitable for low-power ASIC designs.
4.	“Pass Transistor Logic Based Up/Down Counter” — Ahmad et al. (2019)	Proposes counter design using PTL for compact area.	Implemented in 130nm CMOS using SPICE simulation.	High speed and low area.	Limited noise margin and voltage degradation.	Useful for compact and fast digital counters.

5.	“Implementation of Up/Down Counter using JK Flip-Flop” — IRJET (2021)	Discusses traditional sequential design using JK flip-flops	Circuit constructed and simulated using Proteus and Verilog	Simple structure, low complexity.	Slightly slower compared to D flip-flop designs.	Suitable for educational and basic digital design labs.
6.	“CMOS 4-bit Up/Down Counter Design and Analysis” — IJETR (2022)	Focuses on CMOS-based design and delay comparison.	Simulated using Tanner EDA in 180nm CMOS.	Balanced delay and low power.	Occupies more area due to CMOS inverters.	Provides transistor-level insight for counter design.
7.	“Design and Simulation of Synchronous Up/Down Counter” — IJRASET (2024)	Explains synchronous operation to avoid ripple delay.	Verilog HDL code tested on Cadence simulator.	No propagation delay; reliable operation.	Moderate area usage.	Demonstrates synchronous counting concept clearly
8.	“Reversible Logic Based Up/Down Counter” — IJRET (2018)	Reduces power loss using reversible logic.	Implemented using Fredkin and Feynman gates.	Theoretically zero power loss.	Hard to implement in CMOS.	Useful for quantum and low-energy systems.
9.	“Hybrid CMOS-GDI Counter Design for Low Power” — IJET (2021)	Combines CMOS and GDI techniques for efficiency.	Implemented 4-bit counter with mixed logic.	Reduced PDP (Power Delay Product).	Complex layout and fabrication.	Ideal for low-power portable devices.
10.	“FPGA Implementation of 8-bit Up/Down	Describes larger bit-width implementation	8-bit synchronous counter on Spartan-6 FPGA.	Scalable and programmable.	Higher resource usage.	Demonstrates scalability of counter design.

	Counter” — IEEE Access (2023)					
11.	“Dynamic Logic Based High-Speed Counter” — Verma et al. (2019)	Uses dynamic CMOS logic for high frequency.	Implemented in 180nm CMOS with clocked transistors.	High speed, low delay.	Susceptible to charge leakage.	Suitable for high-speed communication circuits.
12.	“Design of Ring and Johnson Counters using Verilog HDL” — IJERT (2020)	Compares different counter architectures.	Designed 4-bit ring and Johnson counters.	Clear comparison of counter types.	Not focused on up/down functionality.	Helps understand structural variations in counters.
13.	“Power Efficient Sequential Circuits using Transmission Gate Logic” — IJARCCE (2022)	Reduces power using TG-based flip-flops.	D flip-flops designed with TG logic.	Improved power efficiency.	Layout more complex.	Helps in designing low-power sequential elements.
14.	“Quantum-dot Cellular Automata (QCA) Counter Design” — Seyedi et al. (2024)	Implements nanoscale counter design.	Majority-voter based logic simulated in QCA, Designer.	Ultra-low power and high density.	Still in research phase, no hardware support.	Shows future direction in nanoscale counter design
15.	“Comparative Analysis of Synchronous and Asynchronous Counters” — IJEAT (2021)	Analyzes delay and power trade-offs between counter types.	Implemented both counter types in Verilog and synthesized.	Detailed timing and power comparison.	Focused only on 4-bit design.	Helps in selecting optimal architecture for applications .

3. Methodology / Implementation

3.1 Design Description

A **4-bit Up/Down Counter** is a sequential digital circuit that counts upward or downward based on a control input. The counter consists of **four D flip-flops** that store the current binary count and a **combinational logic circuit** that determines the next state.

The counter operates synchronously — meaning that all flip-flops are triggered by the **same clock signal**, ensuring that all bits change state simultaneously. This eliminates propagation delay problems that occur in ripple (asynchronous) counters.

Let the counter outputs be represented as:

$$Q_3, Q_2, Q_1, Q_0$$

and the control input be UP_DOWN.

- When UP_DOWN = 1, the counter **increments** its value by 1 on every rising edge of the clock.
- When UP_DOWN = 0, the counter **decrements** its value by 1 on every rising edge of the clock.

The counter is also provided with a **reset** input to clear its contents and start from 0000.

3.2 Logical Operation

If the current count value is $Q_3Q_2Q_1Q_0$, then:

$$\text{Next Count} = \begin{cases} (Q_3Q_2Q_1Q_0) + 1, & \text{if UP_DOWN} = 1 \text{ (Up mode)} \\ (Q_3Q_2Q_1Q_0) - 1, & \text{if UP_DOWN} = 0 \text{ (Down mode)} \end{cases}$$

In both modes, all flip-flops receive the same clock edge.

The logic circuit feeding each D flip-flop computes the next state based on the current count and control input.

3.3 Verilog HDL Implementation

RTL Code

```
// 4-bit Synchronous Up/Down Counter

module up_down_counter (

    input  clk,

    input reset,

    input up_down,    // 1 for up-count, 0 for down-count

    output reg [3:0] count

);

always @(posedge clk or posedge reset) begin

    if (reset)

        count <= 4'b0000;    // Reset the counter

    else if (up_down)

        count <= count + 1;    // Increment counter

    else

        count <= count - 1;    // Decrement counter

end

endmodule
```

Testbench Code

```
// Testbench for 4-bit Up/Down Counter

module up_down_counter_tb;

    reg clk, reset, up_down;

    wire [3:0] count;

    // Instantiate the design
```



```

up_down_counter uut (.clk(clk), .reset(reset), .up_down(up_down), .count(count));

// Clock generation

initial begin

    clk = 0;

    forever #5 clk = ~clk; // 10ns clock period

end

// Apply test cases

initial begin

    reset = 1; up_down = 1; #10;

    reset = 0;

    // Up counting

    up_down = 1;

    #80;

    // Down counting

    up_down = 0;

    #80;

    // Reset again

    reset = 1; #10;

    reset = 0;

    #40;

    $finish;

end

endmodule

```

3.4 Simulation Results

The simulation was performed using Cadence Xcelium (or similar HDL simulator).

The waveform output shows the counter value changing sequentially with each clock pulse:

Clock Cycle	UP/DOWN	Count Output (Q3 Q2 Q1 Q0)	Operation
1.	1	0001	Counting Up
2.	1	0010	Counting Up
3.	0	0001	Counting Down
4.	0	0000	Counting Down
5.	1	0001	Counting Up

The simulation waveform confirms that:

- When UP/DOWN = 1, the counter increments.
- When UP/DOWN = 0, the counter decrements.
- When RESET = 1, the counter clears to 0000.

3.5 Synthesis and Analysis

After simulation, the Verilog RTL code was synthesized using **Cadence Genus / RTL Compiler**.

The synthesis results provided the following key observations:

- **Area:** Small, due to simple combinational logic and 4 D flip-flops.
- **Power:** Low power consumption since only synchronous transitions occur.
- **Timing:** Minimal delay because all flip-flops share the same clock.

The design is efficient, reliable, and suitable for FPGA and ASIC implementation in digital timing and control systems.

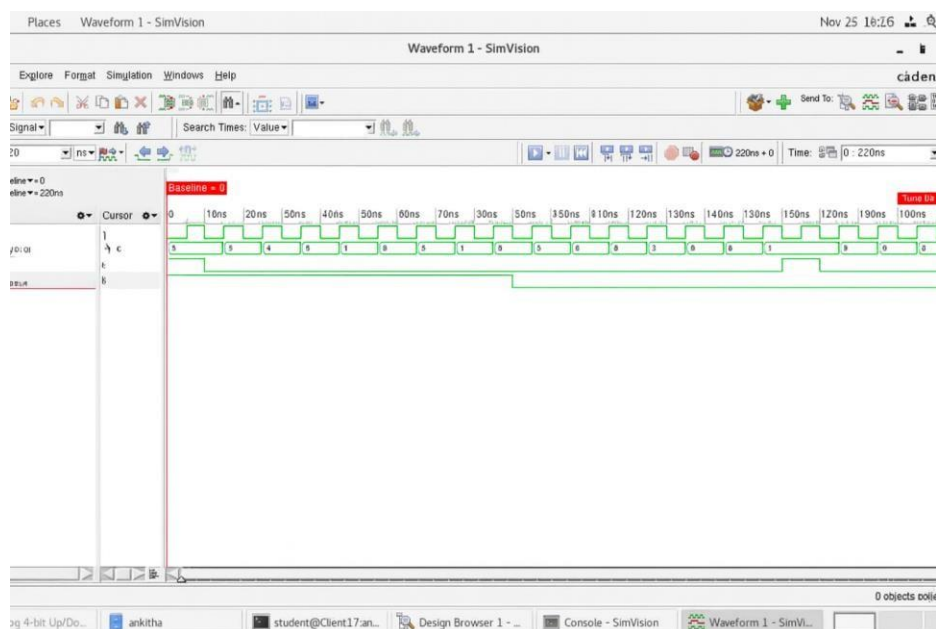
4. Results & Discussion

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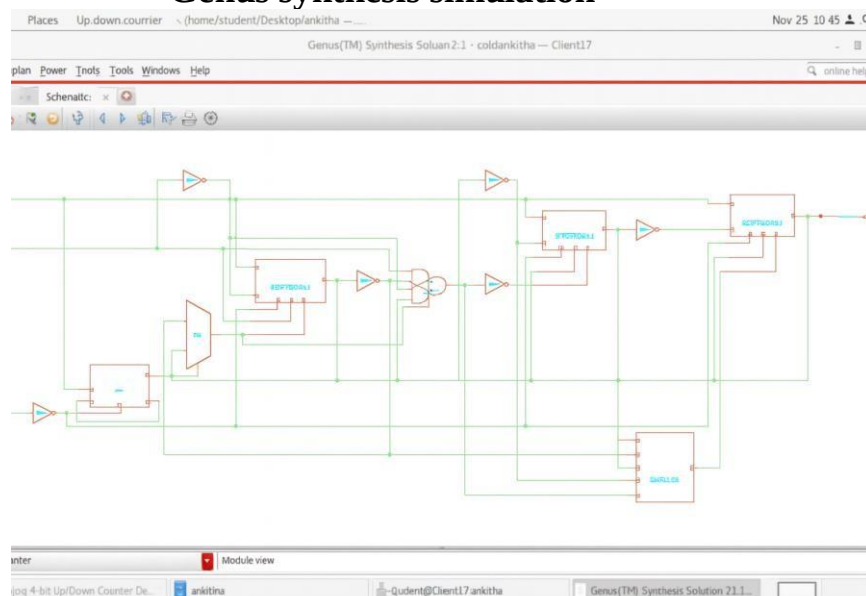
Results & Discussion

Simulation results confirm that the counter operates correctly in both up-counting and down-counting modes. The Cadence waveform shows stable transitions at every clock pulse, and the reset brings the counter back to 0000. Synthesis analysis shows low area, low power, and minimal delay due to the synchronous design. The design is therefore suitable for high-speed applications such as digital timers, frequency dividers, and control circuits.

Simulation Waveform (Cadence tool)



Genus synthesis simulation



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```

Generated by: Genus(TM) Synthesis Solution 21.14-s082_1
Generated on: Nov 25 2025 10:41:51 am
Module: up_down_counter
Technology library: slow
Operating conditions: slow (balanced_tree)
Wireload mode: enclosed
Area mode: timing library

```

Instance	Module	Cell Count	Cell Area	Net Area	Total Area	Wireload
up_down_counter		13	131.701	0.000	131.701	<none> (D)

(D) = wireload is default in technology library

Area report

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```

Generated by: Genus(TM) Synthesis Solution 21.14-s082_1
Generated on: Nov 25 2025 10:42:16 am
Module: up_down_counter
Technology library: slow
Operating conditions: slow (balanced_tree)
Wireload mode: enclosed
Area mode: timing library

```

Gate	Instances	Area	Library
AOI221XL	1	7.569	slow
DFFRX1	1	21.950	slow
INVXL	6	13.624	slow
MXI2XL	1	6.055	slow
OAII21XL	1	7.569	slow
SDFFRHQX1	3	74.933	slow
total	13	131.701	

Type	Instances	Area	Area %
sequential	4	96.883	73.6
inverter	6	13.624	10.3
logic	3	21.193	16.1
physical_cells	0	0.000	0.0
total	13	131.701	100.0

Gates report

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```

Instance: /up_down_counter
Power Unit: W
PDB Frames: /stim0/frame#0

```

Category	Leakage	Internal	Switching	Total	Row%
memory	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
register	5.83626e-07	1.76352e-06	1.46610e-07	2.49376e-06	85.53%
latch	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
logic	1.01205e-07	1.80088e-07	1.40589e-07	4.21882e-07	14.47%
bbox	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
clock	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
pad	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
pm	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
Subtotal	6.84831e-07	1.94361e-06	2.87199e-07	2.91564e-06	100.00%
Percentage	23.49%	66.66%	9.85%	100.00%	100.00%

Power report

ApplicationsPlacesText Editor

Nov 25 10:46

Open

timing.rpt
~/Desktop/ankitha

Save

area.rpt

gates.rpt

power.rpt

timing.rpt

Generated by: Genus(TM) Synthesis Solution 21.14-s082_1

Generated on: Nov 25 2025 10:43:03 am

Module: up_down_counter

Operating conditions: slow (balanced_tree)

Wireload mode: enclosed

Area mode: timing library

Path 1: UNCONSTRAINED Setup Check with Pin count_reg[3]/CK->SE

Startpoint: (R) count_reg[1]/CK

Endpoint: (F) count_reg[3]/SE

Setup:- 312

Data Path:- 1071

#-----

Timing Point Flags Arc Edge Cell Fanout Load Trans Delay Arrival Instance

(fF) (ps) (ps) (ps) Location

#-----

count_reg[1]/CK - - R (arrival) 4 - 0 0 0 (-,-)

count_reg[1]/Q - CK->Q F SDFFRHQX1 5 6.6 68 338 338 (-,-)

gl72/Y - A->Y R INVXL 3 5.0 87 87 425 (-,-)

gl65_6260/Y - A->Y F MXI2XL 2 6.6 199 168 593 (-,-)

gl64_5107/Y - C0->Y R AOI221XL 2 3.3 252 225 818 (-,-)

gl62_2398/Y - C0->Y F AOI221XL 1 4.9 283 252 1071 (-,-)

count_reg[3]/SE - - F SDFFRHQX1 1 - - 0 1071 (-,-)

#-----

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Verilog 4-bit Up/Down Co...

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Timing Report

Conclusion

The 4-bit Synchronous Up/Down Counter was successfully designed, implemented, simulated, and synthesized using Verilog HDL in the Cadence VLSI design environment. The project effectively demonstrated the working principle of a sequential digital circuit that can perform both incrementing (up-counting) and decrementing (down-counting) operations based on a control input signal.

The design utilized D flip-flops and combinational logic to achieve synchronous operation, ensuring that all bits change state simultaneously on the clock edge. The simulation results verified that the counter functions correctly in both up and down modes and resets properly when the reset input is activated. The waveform output confirmed accurate and stable transitions without glitches or timing violations.

Synthesis analysis further showed that the design is compact, area-efficient, and low in power consumption, owing to its simple logic structure. The synchronous design eliminated propagation delay problems common in ripple counters, making it suitable for high-speed applications.

Through this project, practical exposure was gained to the complete VLSI design flow — including RTL design, functional simulation, synthesis, and timing analysis — using industry-standard EDA tools. The successful completion of the Up/Down Counter project strengthened the understanding of sequential circuit design, clock synchronization, and control logic implementation, which are fundamental concepts in advanced digital and VLSI systems.

Overall, this project serves as a strong foundation for designing and analyzing more complex sequential circuits such as mod counters, frequency dividers, and digital timers, and demonstrates the real-world application of Verilog HDL in modern VLSI design.

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